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INITIAL SYSTEM PROPOSAL

KADA COOPERATIVE SYSTEM

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PROPOSED SYSTEM BACKGROUND

1. Introduction

KADA is the primary agency committed to improving the living standards of farmers in the Kemubu region of Malaysia [1]. Over the years, KADA has managed its loan and membership applications through manual processes, which only use paper form and need to insert data manually. This has proven to be inefficient, time-consuming and environmentally unfriendly. These outdated methods have resulted in issues such as record loss and delays in handling applications. As a way of addressing these issues, KADA has proposed the development of an innovative solution which is a website application named the “KADA Cooperative System” specifically to increase accessibility and achieve efficiency in the operations of membership and loan applications.

The objectives of this project include:

- i. To learn the practical application of SDLC phases including planning, analysis, design, implementation and testing phases.
- ii. To modernize and digitalize KADA’s manual processes for loan and membership applications.
- iii. To evaluate and design the key features of the proposed system, such as membership registration, loan processing and reporting to the BODs.
- iv. To focus on system elements like database management, user interface design and any reliable performance.
- v. To execute testing methodologies for ensuring the smooth system, error corrections and maintain performance in the processes.

2. The Existing System

The current system of the KADA Cooperative is a manual system that relies heavily on paper-based processes. It is used to perform the following functions:

1. Membership Registration

Applicants fill out a paper-based membership form to provide their data, which is then registered to make them members of the cooperative.

2. Loan Applications

Members use a manual loan application form on paper to apply for loans from the cooperative.

3. Financial Statements for Members

Members are provided with paper-based financial statements that detail their current financial status within the KADA Cooperative, including both savings and loans.

4. Data Storage

Data collected on paper is manually transferred into a computerized system for storage.

5. Financial Reporting

Financial data is manually calculated and compiled into reports, which are presented to the board of directors on a monthly and yearly basis.

The problems faced by the client of the system

1. The non-tracking of online loan status

The members of this cooperative are unable to track the status of their loan applications online. This makes them visit the cooperative office or call up the staff to find out about the status of their loan applications. Such a method of tracking creates inefficiencies and causes delays in communication.

2. Limited access to information on dividends

Dividends are distributed only once a year according to the profit of the cooperative. This means that members have no real-time access to information about how dividends are computed and whether they are eligible for it, thus needing to follow up manually to keep informed.

3. Physical storage of records

Documents from the types of forms are stored physically and not in any online storage system, wherein each factor increases the risk of losing, damaging, or misplacing documents. It also limits the access provided and makes data retrieval a very tedious process.

4. Inconvenient access to savings and reports

They don't have access to their savings and annual reports online. Thus, members have to visit the cooperative office, waiting and spending time, especially for those from a distance.

5. No direct framework to identity or address fraud

There are no solid mechanisms in order to ascertain fraudulent actions of a person making false income declarations. The manual system does not compare and validate member information to determine whether doubt exists, and no fraud cases have been reported.

3. The Proposed System

Mission Statement

To have a fully integrated digital platform for KADA's cooperative system that provides an easy and hassle-free management of membership and loan cases, with security for data handling and user experience, allowing transparency, efficiency, and ecological operational activities for all the users.

Overview of the System

The KADA Cooperative System is an online system that automates all processes from the cooperative levels by offering the most effective way of regulating membership applications, loan applications, data management, and reporting. This tool excludes manual activities by providing users with a centralized, accessible, and secure system. The system will serve four key actors: applicants, cooperative members, administrators, and the Board of Directors. Applicants will create an account and check on membership status, while members will apply for loans and view financial information. Administrators will ensure that operations within the system run smoothly and applications are approved to be processed. The BOD will make informed decisions by comparing analytics at high levels and reports from the cooperatives' performances.

Objective Statement

This proposal describes the KADA Cooperative System, an integrated digital platform in response to the requirements of KADA, aimed at enhancing the overall functioning of the cooperative and at improving user experience. The main goal is to simplify the procedure of membership and loan applications with secure, effective data management among the stakeholders.

The idea is to develop a web-based application that can be accessed through PCs and other mobile-friendly interfaces. It shall be used by applicants, members, administrators, and the Board of Directors, and it will have different functions for each. Administrators will especially be given the facility to process applications, verify documents, and prepare reports, whereas for the members, their applications and financial statements will automatically be updated.

Additionally, this platform will securely allow the Board of Directors to access these platforms for efficient analysis in implementing strategic decisions on the performances of the cooperatives. One of the key features will involve embedding an integrated verification system into the platform for authenticating membership and loan applications to inspire trust and added transparency.

The proposed system will further ease data storage and management by enabling KADA to manage member and loan records with ease. All the user accounts will link to a single database, allowing the automatic updating of data when new information is put in. This will be an environmentally friendly approach, reducing manual processes and use of paper for records, hence being more efficient.

In a nutshell, our aim is to implement the robustness and scalability of the system that enhances users' experience, as well as complementing sustainable growth objectives of KADA. We intend to use tools such as PHP, HTML, Bootstrap, and MySQL in implementing this project successfully. Upon the close of this project, our hope is that all the objectives stated will be complied with to meet the successful deployment of the KADA Cooperative System.

System Scope And Boundaries

KADA Cooperative System is a digital platform designed to enhance its current membership and loan application procedure, as well as improve user experience, data security, and operational efficiency. The system is designed for four major user types: applicant, member, administrator, and the Board of Directors (BOD). Each user type has specific defined duties inside the system and what interaction is possible with the system, thereby leading to a specified, role-specific user experience.

However, for applicants, it will allow the chance to send membership applications remotely and receive real-time updates on their application status. Once they have been approved, applicants then become members with additional functions such as loan applications, financial statement viewing, and monitoring of loan status. Further, the system generates statements for members, which provide insights into their general financial activities and commitments.

On the other side, administrators are responsible for managing things such as approval of the membership or loan application, maintenance of financial records, and monitoring of user accounts. They also perform system administration tasks such as ensuring a precise database, preparing financial reports to the BOD, and facilitating member-cooperative communication.

The technology is generally used by the Board of Directors for senior agency functions, application approval, financial report analysis, and access to BOD-reporting features specific to membership and loan information. All these aspects are meant to facilitate the BOD in making decisions that are reasonable and informative regarding the cooperative's smooth running.

Within the system boundaries, they are mere functionalities that need to be kept simple. This financial system deals only with external functions that are not controlled by the system, such as payment processing. The payments are tracked by the system, but the actual transaction occurs externally. Authentication of someone's identity through personal interaction is not a

part of the system's responsibilities.

The system, as a whole, would store user, membership, and loan data in a common database in a safe and secure manner. The whole system is ready to be used by developing a web framework, which can be used in many devices and later add new features according to the need. Notifications such as updates or alerts could be delivered through text messages, emails, etc.

Currently, this system allows application for membership, examination of financial statements, application for loan, assessment of applications, generation of reports, and administration activities, while showing clearly its boundaries on what is included in its functionalities as well as the aspects excluded outside the system, such as payment integrations and human verification methods.

This is intended to provide such a strong platform that is easy to use and function well for KADA while still able to update non-technical stakeholders and technical evaluators.

Use Case Diagram

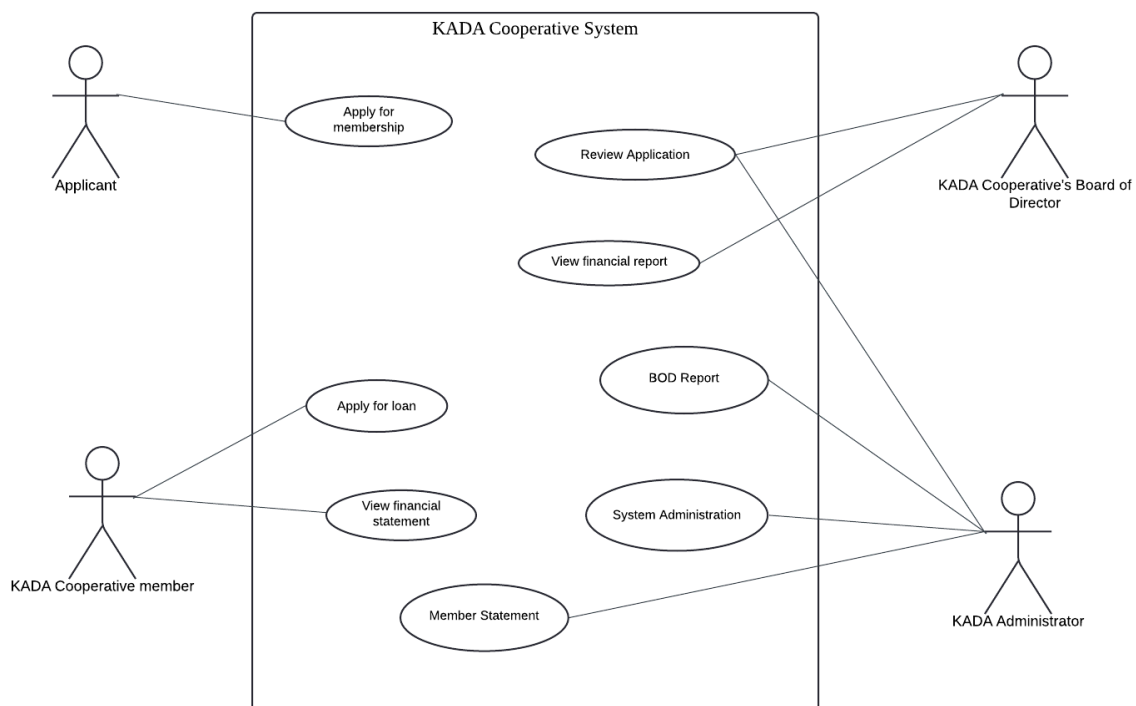


Figure 1 : Use Case of the proposed system

High quality image view: [UC Diagram: Lucidchart](#) Version 1.1

Proposed Technology

We will use a web app to implement our system. In such a case, this will be provided access for users anywhere using a simple web browser without installing software on their devices. The web application framework eliminates compatibility problems across other operating systems like Windows or MacOS. Likewise, the web development architecture guarantees multiple simultaneous users without losing communication and functional activities among all, which also includes members, administrators, and the Board of Directors (BOD).

To build a user-friendly interface for the application, the front part of the system will be developed using HTML, CSS and a responsive framework like Bootstrap. HTML will define the structure and contents of the web pages while CSS and Bootstrap will style them to ensure the pages look neat and function well across multiple platforms such as desktops and mobile phones. These technologies will ensure that the interface of the system is friendly for everyone regardless of the technical background of the user.

The backend of the system will be entirely made of Django, which is a Python-based framework for web application development. It will manage all the server activities associated with user input validation for session data management and database security. Security features built into Django would keep secure sensitive data, including member information and financial data.

The system will be capable of managing and storing data using MySQL, the relational database management system. It is a highly trusted open-source database for the efficient and effective storing and retrieving of data. It is organizing the data into different structured tables like member details, loan applications, financial records, and so on, while ensuring the integrity and consistency of the data through its constraints and linkages. It is going to allow scalability along with the users and the transactions in course of time.

Major communication technologies in Malaysia will be implemented as part of the system, including WhatsApp and Google Mail (Gmail). WhatsApp will be used in sending urgent notifications like follow-ups about loan applications, membership approvals, and reminders right to users' smartphones. More importantly, Google Mail will be at stake for sending such lengthy email alerts or formal documents to subscribers to ensure clear and effective communications. Since these are common tools, the engagement for terms and updates would be in time.

Thus, our system would provide a convenient, reliable, and safe experience for all stakeholders using these technology systems. This combination of a friendly front end, strong back-end architecture, and efficient database meets the needs of the KADA Cooperative in an effective and efficient manner.

4. Glossary

This document use the following terms and abbreviations:

Term	Descriptions
KADA	Kemubu Agricultural Development Authority
SDLC	System Development Life Cycle
BOD	Board of Director
PC	Personal Computer
HTML	Hypertext Markup Language
PHP	HyperText Preprocessor
MYSQL	My Structured Query Language
GMAIL	Google Mail
CSS	Cascading Style Sheet

5. Bibliography

[1] *Portal Lembaga Kemajuan Pertanian Kemubu | KADA KEMUBU*. (n.d.).

<http://www.kada.gov.my/>